higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

2 DECEMBER 2019

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This marking guideline consists of 7 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	C		
	1.1.2	A		
	1.1.3	11		
	1.1.4	A		
	1.1.5	B		
			(5 × 1)	(5)
1.2	1.2.1	D		
	1.2.2	G		
	1.2.3	I		
	1.2.4	C		
	1.2.5	F		
			(5 × 1)	(5)
1.3	1.3.1	volumetric		(1)
	1.3.2	(a) voltage (b) velocity		(2)
	1.3.3	differential pressure square root		(2)
				[15]

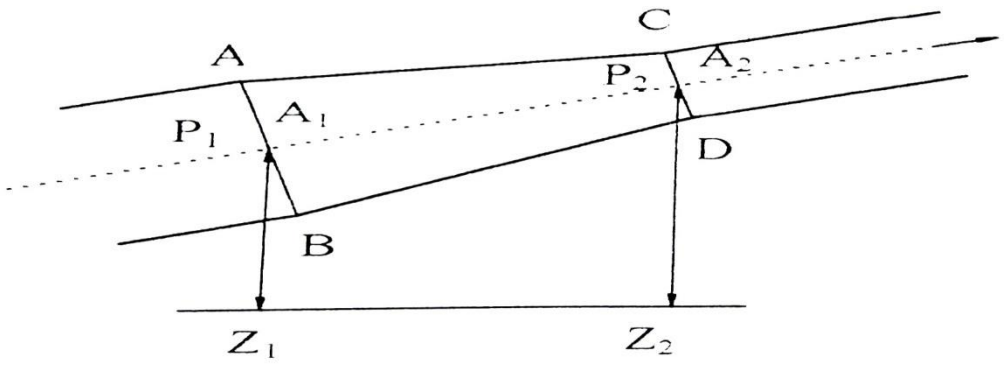
TOTAL SECTION A: 15

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SECTION B

QUESTION 2

2.1



A fluid stream constricted from area A_1 at AB to a smaller area A_2 at CD

- The fluid at first is assumed to be incompressible, so that the density is unchanged from section AB to CD.✓
- From Bernoulli's theory ,we have that total energy across AB = total energy across CD✓
- Thus $P_1V_1 + \frac{1}{2}m{v_1}^2 + mgZ_1 = P_2V_2 + \frac{1}{2}m{v_2}^2 + mgZ_2$ (1)✓ where P_1 and P_2 are pressures across AB and CD respectively. Also V_1 and V_2 are volumes of liquid across AB and CD respectively
- For incompressible fluid , V_1=V_2=V(say)
- Now $V = \frac{m}{\rho}$……(2) ✓

We can rewrite equation (1) as follows:

- $P_1\frac{m}{\rho} + \frac{1}{2}m{v_1}^2 + mgZ_1 = P_2\frac{m}{\rho} + \frac{1}{2}m{v_2}^2 + mgZ_2$ ……(3) ✓

Divide equation (3) by mg gives

- $\frac{P_1}{\rho g} + \frac{{v_1}^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{{v_2}^2}{2g} + Z_2$……(4)✓

Manipulating equation (4) gives ;

- $\frac{{v_2}^2}{2g} - \frac{{v_1}^2}{2g} = \left[\frac{P_1}{\rho g} + Z_1\right] - \left[\frac{P_2}{\rho g} + Z_2\right]$……..(5)✓

Pressure at AB with the height taken into consideration, will be higher than the pressure at CD. The difference in pressure with height included (h) = pressure at AB – pressure at CD

- That is $h = \left[\frac{P_1}{\rho g} + Z_1\right] - \left[\frac{P_2}{\rho g} + Z_2\right]$…….(6)✓

Substitute (6) into (5) and simplify ,we have

- ${v_2}^2 - {v_1}^2 = 2gh$……..(7)✓
- Continuity of flow demands that the quantity of fluid, Q, flowing through section AB is the same as that through section CD.

Now since Q is the product of area and velocity at each section it implies that

- $Q_{AB} = Q_{CD} \Rightarrow A_1v_1 = A_2v_2$ ……..(8) ✓
- $\Rightarrow mv_2 = v_1$ ……(9),✓

where $m = \frac{A_2}{A_1}$, and v_1 , v_2 are velocities at AB and CD respectively

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Substituting (9) into (7) and simplifying, we get:

$$v_2{}^2 - [mv_2]^2 = 2gh$$
$$\Rightarrow v_2{}^2 = \frac{2gh}{1-m^2}\checkmark$$
$$\Rightarrow v_2 = \frac{1}{\sqrt{1-m^2}}\sqrt{2gh}$$
$$\Rightarrow v_2 = E\sqrt{2gh} \quad \text{.....(10)}\checkmark$$
$$\text{where } E = \frac{1}{\sqrt{1-m^2}}$$

Thus from equation (8) it follows that $Q = A_2 v_2 = A_2 E\sqrt{2gh}\checkmark$ (15)

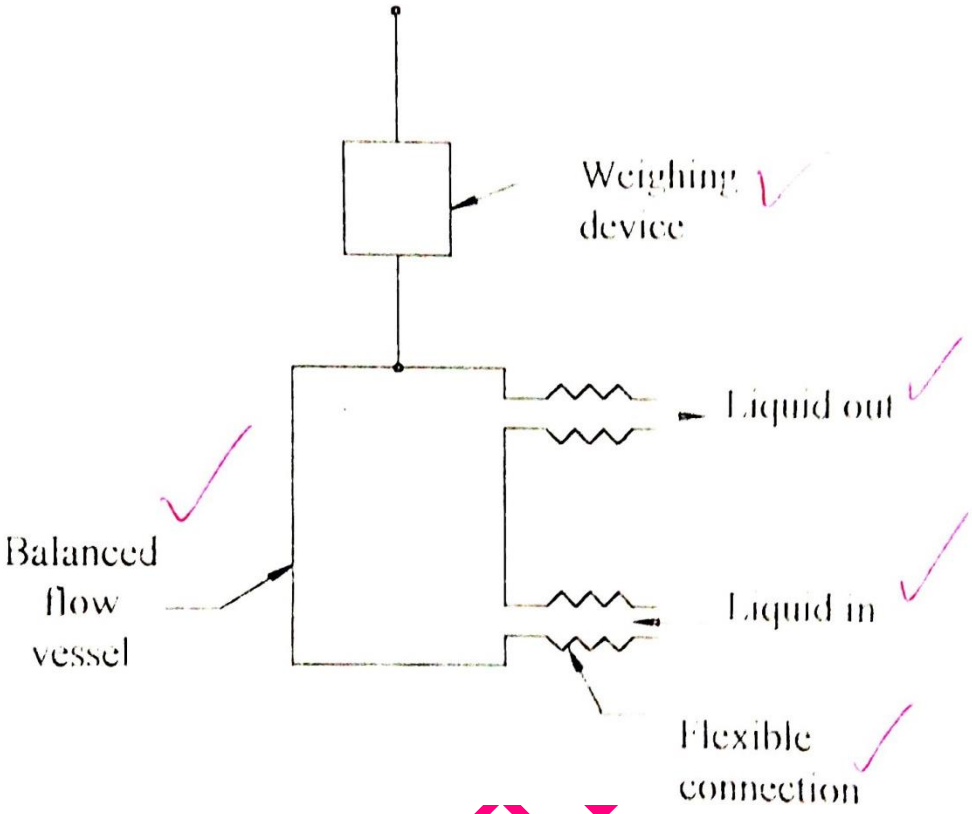
2.2
- Discharge coefficient,✓ C–A correction factor for the friction effects and another for the contraction effect.✓
- Expansion factor✓ – Due to temperature changes in a flow line, this factor must be taken into consideration✓
- Reynolds pipe size factor✓ – It correct all the changes in pipe size due to flow✓ (6)

2.3
$$Q = \frac{8}{15}tan\left(\frac{\theta}{2}\right)\sqrt{2gH^5}$$
$$0{,}7 = \frac{8}{15}tan\left(\frac{50}{2}\right)\sqrt{2 \times 10 \times H^5}$$
$$H^5 = \frac{1}{20} \times \left[\frac{0{,}7 \times 15}{8tan\left(\frac{50}{2}\right)}\right]^2$$
$$H = \left[\frac{1}{20} \times \left[\frac{0{,}7 \times 15}{8tan\left(\frac{50}{2}\right)}\right]^2\right]^{\frac{1}{5}}$$
$$H = 0{,}831\ m$$
(4)
[25]

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QUESTION 3

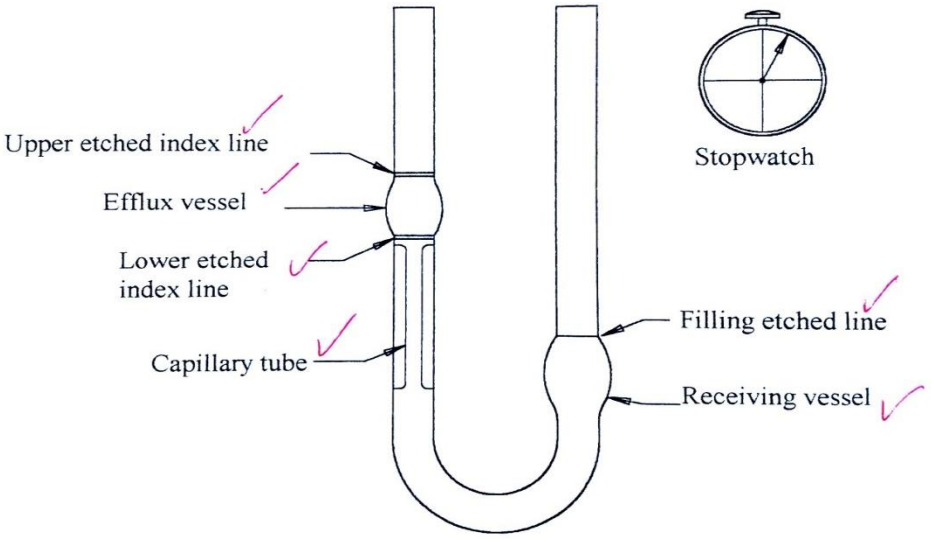
3.1



(5)

3.2 A fixed volume vessel is employed,✓ through which the measured liquid flows continuously.✓ The vessel is weighed automatically by a scale, spring balance or pneumatic force balance transmitter.✓ Since the weight of a definite volume of a liquid is known,✓ the instrument can be calibrated to read directly in specific gravity or density units✓ (5)

3.3



The capillary tube viscometer consists of a fluid reservoir, to hold a specific volume of sample liquid and a capillary tube.✓ The hydrostatic head of fluid causes the liquid to flow through the capillary.✓ A sample liquid of fixed volume is charged to the lower receiving vessel✓ and the viscometer is placed in a themostatic bath.✓ After time is allowed for the sample liquid to reach thermal equilibruim,✓ the sample is drawn up into the efflux vessel by suction until the level is above upper etched index line.✓ The fluid is the allowed to flow down through the capillary by releasing the suction.✓ When the fluid surface passes the upper etched index line, a stopwatch is started.✓ The watch is stopped when the surface passes the lower etched index line of the efflux vessel.✓ From this time, the kinematic viscosity of the fluid is calculated by multiplying it by the viscometer calibration constant.✓ (15)

[25]

QUESTION 4

4.1 A screened glass electrode can be used and the electrode assembly must be located at a point where the residual field is smallest. (2)

4.2 Screening is necessary for the insulated cable attached to the glass electrode. (1 × 2) (2)

4.3 The solution is neutral.✓ That means that the concentration of hydrogen ion equals the concentration of hydroxyl ions in a solution. ✓ (2)

4.4 To provide automatic temperature compensation, it involves a property of the zero shift and slope shift voltages✓ to be equal and opposite.✓ This point on the pH scale is known as isopotential point✓. (3)

4.5
- Washing with water and wiping with cotton wool soaked in a diluted hydrochloric acid will suffice.
- In severe cases, the electrodes may require soaking in solvents.
- Inorganic solvents are preferable for greasy and oily deposits.
- Organic solvents have a dehydrating effect on the membrane and therefore only brief immersion in a solvent followed by soaking in hydrochloric acid should be used.
- A jet of solvent may be used periodically for frequent cleaning.
- Ultrasonic transducer attached to a thin plate may be used. (6)

[15]

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QUESTION 5

5.1 5.1.1 (a) Control element
(b) three or less (2)

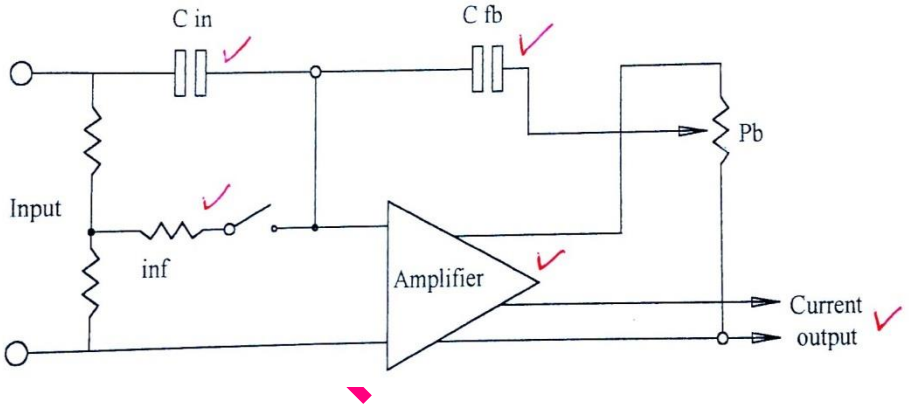
5.1.2 (a) inactive neutral
(b) last position
(c) slightly beyond (3)

5.1.3 equilibrium (1)

5.1.4 percentage (1)

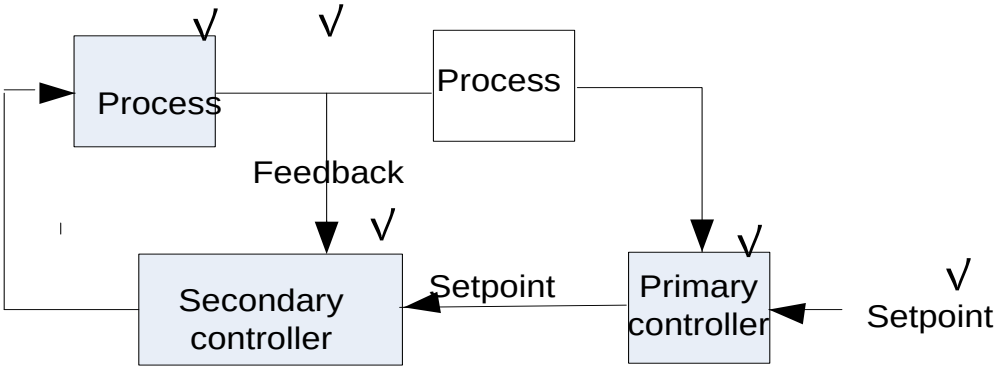
5.1.5 (a) process variable
(b) desired value
(c) corrective signal
(d) pre-set (4)

5.2



(5)

5.3



(4)
[20]

TOTALSECTION B: 85
GRAND TOTAL: 100

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